ALEKYA MADDU

SENIOR FULL STACK SOFTWARE DEVELOPER — JAVA / SPRING / REACT

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PROFESSIONAL SUMMARY

Senior Full Stack Software Developer with 6+ years architecting secure, scalable enterprise applications across Banking and Healthcare using React.js 19, TypeScript, Next.js, Java, Spring Boot, Spring Security, and Microservices. Hands-on experience hardening applications with Spring Security (JWT, OAuth 2.0, RBAC), decomposing monoliths into Docker/Kubernetes-based microservices. Early adopter of Spring AI for embedding LLM-assisted capabilities into Java services. Delivered measurable outcomes: 40% API latency reduction, WCAG 2.1 AA compliance across enterprise platforms, and reliable CI/CD-driven microservices delivery on AWS and Azure.

WORK EXPERIENCE

Senior Full Stack Software Developer

April 2025 – Present

Collabera / Bank of America | Chicago, IL (Hybrid)

- **Challenge:** HR Connect applications serving 100,000+ bank employees failed WCAG accessibility standards. **Action:** Remediated the platform using semantic HTML5, ARIA, JavaScript, jQuery, JSP, and Bootstrap. **Result:** Achieved WCAG 2.1 AA compliance across enterprise HR experiences.
- **Challenge:** The bank needed to retire a third-party OpenText CMS in favor of an in-house solution. **Action:** Built React/TypeScript proof-of-concepts backed by Spring Boot, Spring Security, Hibernate, Maven, and PostgreSQL microservices, collaborating with UX, QA, and backend teams. **Result:** Delivered a validated microservices-based CMS architecture that became the foundation for enterprise HR application modernization.
- **Action:** Secured the new CMS microservices by implementing Spring Security with JWT-based authentication, method-level authorization, and CSRF protection, replacing the legacy session-based auth model in OpenText. **Result:** Strengthened the application's security posture ahead of enterprise rollout.
- **Action:** Led migration of shared UI components from Bootstrap 4.x to Bootstrap 5.x across enterprise HR modules — removing jQuery-dependent plugins, adopting utility-first classes, and updating custom themes/breakpoints. **Result:** Improved cross-browser consistency and cut front-end technical debt ahead of the CMS cutover.
- **Action:** Applied React lazy loading, code splitting, responsive design, and reusable components to optimize data structures. **Result:** Reduced page load times and improved responsiveness across enterprise HR applications.
- **Action:** Prototyped Spring AI within Spring Boot microservices to explore AI-assisted content tagging and search recommendations for the HR CMS, evaluating LLM providers for enterprise content-retrieval use cases. **Result:** Produced a working proof-of-concept demonstrating Spring AI's fit for enterprise content workflows.
- **Action:** Partnered with QA to validate screen-reader compatibility using JAWS, NVDA, and VoiceOver, and published 200+ WCAG-compliant updates in OpenText CMS with offshore development teams. **Result:** Maintained accessibility standards across enterprise HR applications throughout the transition.

Software Developer

February 2023 – February 2025

Abbott Laboratories | Illinois

- **Action:** Developed and enhanced a large-scale Healthcare Management System using React, Redux, Java, Spring Boot, Spring Security, MongoDB, and REST APIs supporting patient care, medical records, and provider communication. **Result:** Delivered HIPAA-compliant workflows relied on by clinical and administrative staff.
- **Action:** Integrated Spring Boot RESTful APIs with React and built real-time messaging with WebSockets. **Result:** Cut data retrieval latency by 40% and improved provider communication efficiency by 40%.
- **Action:** Built scheduling, calendar, and patient dashboard modules using React Hooks, Redux, and Next.js SSR. **Result:** Reduced appointment booking errors by 20% and improved application usability.
- **Challenge:** A monolithic architecture limited scalability and release velocity. **Action:** Contributed to decomposing it into Docker-based microservices deployed on Azure Kubernetes Service (AKS), alongside 30+ optimized SQL queries, stored procedures, and triggers. **Result:** Improved scalability and maintainability while cutting query response times by 25%.
- **Action:** Implemented the Spring Security framework — JWT Authentication, OAuth 2.0, and Role-Based Access Control (RBAC) — to secure patient, provider, and administrative workflows. **Result:** Supported HIPAA-compliant access control across the platform.

- **Action:** Integrated Jest, React Testing Library, JUnit, and Log4j for testing, error tracking, and monitoring, using Microsoft Copilot to accelerate development and surface performance bottlenecks. **Result:** Improved release quality and reduced time spent diagnosing production issues.
- **Action:** Built and maintained MERN stack (MongoDB, Express.js, React, Node.js) services for the Healthcare Management System, designing Express.js REST APIs and MongoDB schemas consumed by React front-end modules. **Result:** Streamlined end-to-end data flow between the React UI and MongoDB, speeding up delivery of new patient-facing features.
- **Action:** Optimized MongoDB aggregation pipelines and indexing strategy within the MERN stack, and refactored Node.js/Express.js middleware for request validation and error handling across patient and provider services. **Result:** Reduced average API response times and improved reliability of data-heavy MongoDB queries.
- **Action:** Wrote Python scripts to automate data validation, migration, and reporting jobs across the platform's MongoDB and SQL data stores. **Result:** Cut manual data-processing effort and improved accuracy of downstream operational reports.

Software Developer *March 2019 – December 2021*

Capgemini Technology Services | India

- **Action:** Redesigned the Corporate Customer Onboarding System UI using Angular 10/12, TypeScript, HTML5, and CSS3. **Result:** Delivered responsive, cross-platform experiences aligned to business requirements.
- **Action:** Built D3.js data visualizations and integrated the Angular frontend with Java/SQL backend microservice-style APIs. **Result:** Enabled real-time data synchronization across product features.
- **Action:** Implemented lazy loading, code splitting, and reusable Angular components. **Result:** Reduced initial page-load times by 2.5 seconds, improving overall site performance.
- **Action:** Collaborated with UX designers using Zeplin to translate wireframes into pixel-perfect responsive interfaces. **Result:** Supported consistent, high-fidelity product feature delivery.
- **Action:** Drove Automated Testing, Integration Testing, and TDD using Jasmine, Karma, and Protractor, applying Root Cause Analysis to defects. **Result:** Strengthened QA coverage throughout the SDLC.

PROJECTS

Appointment Booking System | React, Java, Spring Boot, REST APIs, MySQL, Axios

- Built a responsive React front end using Axios to call Spring Boot REST APIs backed by a MySQL database, covering end-to-end appointment data flow from browser to persistence layer.

Student Enrollment Management System | Python, Flask, JavaScript, HTML5, Bootstrap, MongoDB

- Developed a full-stack course enrollment app with a Python/Flask backend, a Bootstrap-based JavaScript/HTML frontend, and MongoDB for NoSQL data storage.

TECHNICAL SKILLS

Frontend: React.js 19, React Hooks, Redux, Next.js, React Router 7.x, Angular (7/10/12), TypeScript, JavaScript (ES6+), HTML5, CSS3, Bootstrap 4.x/5.x (migration experience), Responsive Design, AJAX, JSON, Webpack, Babel, D3.js, Highcharts, Axios, EcraUI

Backend & AI: Java, Spring Boot, Spring MVC, Spring Security, Spring AI (LLM integration, emerging), Hibernate, Maven, Microservices Architecture, REST APIs, GraphQL, Node.js, Express.js, Python, Flask

Programming: Java, JavaScript (ES6+), TypeScript, Python, SQL, Data Structures

Databases: PostgreSQL, MySQL, MongoDB, Firebase, SQL, Stored Procedures, Triggers

Cloud & DevOps: AWS (EC2, S3, Lambda), Microsoft Azure, Docker, Kubernetes (AKS), Jenkins, CI/CD, Git, GitHub, Bitbucket, Linux, WinSCP

Testing & Quality: JUnit, Jest, React Testing Library, Enzyme, Jasmine, Karma, Protractor, Automated Testing, Integration Testing, TDD, Log4j, Quality Assurance

Security & Accessibility: Spring Security, JWT, OAuth 2.0, RBAC, WCAG 2.1 AA, Section 508, ARIA

Tools & Methodologies: Jira, Confluence, Figma, Adobe XD, Zeplin, Axure RP, Postman, VS Code, IntelliJ IDEA, Microsoft Copilot, OpenText CMS, Agile, Scrum, SDLC, Cross-Functional Collaboration, Root Cause Analysis

EDUCATION

University of Central Missouri **January 2022 – May 2023**
M.S., Computer Science *Warrensburg, MO*

Vasireddy Venkatadri Institute of Technology **September 2014 – April 2018**
Bachelor's, Electrical Engineering *Nambur, India*